

SATENAS, Sweden

WMO: 025200

Lat: 58.436N

Lon: 12.707E

Elev: 177

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.8	12.2	3.5	6.6	8.1	8.5	8.5	14.0	30.1	41.1	27.6	39.6	3.8	60	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	78.2	65.4	75.1	64.3	72.0	62.5	67.5	74.6	65.7	72.4	64.0	70.0	9.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.1	70.6	63.2	87.3	68.7	61.5	82.1	66.9	32.0	74.8	30.6	72.6	29.3	70.1	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.5	23.4	20.7	DB	0.2	83.5	7.4	3.1	-5.1	85.7	-9.4	87.6	-13.5	89.3	-18.9	91.5	(4)
(5)			WB	-0.3	70.6	7.3	2.0	-5.6	72.1	-9.9	73.2	-13.9	74.3	-19.2	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	29.7	29.7	34.2	42.8	51.0	57.5	62.0	60.9	54.2	45.6	38.0	32.5	(6)	
	DBStd	13.23	8.19	8.42	7.07	5.84	5.87	4.69	4.47	4.46	4.93	6.05	6.48	8.40	(7)	
	HDD50	3054	629	567	489	228	57	2	0	0	18	160	360	544	(8)	
	HDD65	7369	1094	987	954	666	435	230	118	141	325	601	809	1009	(9)	
	CDD50	1203	0	0	0	11	88	227	371	339	143	23	1	0	(10)	
	CDD65	45	0	0	0	0	1	5	24	14	1	0	0	0	(11)	
	CDH74	369	0	0	0	0	13	61	202	90	3	0	0	0	(12)	
	CDH80	32	0	0	0	0	0	7	22	3	0	0	0	0	(13)	
(14) Wind	WSAvg	9.7	10.5	10.3	10.1	9.6	9.4	9.1	8.7	8.7	9.7	9.9	10.2	10.5	(14)	
(15) Precipitation	PrecAvg	28.1	2.2	1.7	1.4	1.7	1.9	3.0	2.9	2.8	2.7	2.9	2.4	2.4	(15)	
	PrecMax	38.2	4.5	4.6	3.0	3.7	3.6	6.1	6.1	5.9	7.2	6.1	4.8	5.0	(16)	
	PrecMin	16.3	0.4	0.2	0.4	0.2	0.2	0.6	0.1	0.5	0.3	0.4	0.8	0.7	(17)	
	PrecStd	4.8	1.0	1.0	0.7	1.1	0.9	1.3	1.5	1.2	1.6	1.3	1.1	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	46.4	53.5	67.0	76.0	80.2	82.6	79.9	73.2	62.0	53.8	49.3	(19)	
		MCWB	44.5	44.0	45.9	54.2	61.8	64.6	67.7	65.8	62.7	57.4	52.2	48.2	(20)	
	2%	DB	43.7	43.9	49.2	60.9	70.4	74.9	79.1	76.6	68.4	58.0	51.2	46.9	(21)	
		MCWB	42.2	42.1	43.6	51.2	58.4	62.8	66.1	65.3	60.4	55.2	49.6	45.7	(22)	
	5%	DB	41.7	42.1	46.0	56.4	65.8	71.0	76.1	73.4	64.9	56.1	49.4	45.3	(23)	
		MCWB	40.4	40.6	41.7	48.6	56.4	60.5	65.2	63.9	59.2	53.8	48.0	44.0	(24)	
	10%	DB	39.9	40.4	43.3	52.8	61.7	67.4	72.6	70.1	62.3	54.4	47.4	43.2	(25)	
		MCWB	38.7	39.0	40.3	46.5	53.9	58.9	63.7	62.8	57.9	52.1	45.9	42.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.2	44.8	47.3	56.2	64.1	67.2	70.6	69.2	64.8	58.3	52.7	48.5	(27)	
		MCDB	46.0	45.9	51.3	63.8	72.3	75.4	78.3	75.5	69.4	61.2	53.6	48.9	(28)	
	2%	WB	42.7	42.6	44.6	52.1	60.2	64.3	68.3	66.8	62.0	55.9	49.9	46.2	(29)	
		MCDB	43.4	43.4	48.0	59.0	67.6	72.6	76.0	72.9	66.4	57.4	50.8	46.9	(30)	
	5%	WB	40.6	40.9	42.6	49.7	57.5	62.1	66.5	65.2	60.0	54.3	48.1	44.2	(31)	
		MCDB	41.4	41.8	45.1	55.1	63.5	68.8	73.7	71.1	63.8	55.8	49.2	45.0	(32)	
	10%	WB	38.9	39.1	40.8	47.5	55.2	60.1	64.6	63.6	58.4	52.5	46.1	42.1	(33)	
		MCDB	39.8	40.4	43.0	51.8	60.7	66.2	70.8	68.7	61.8	54.1	47.3	43.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.6	11.4	14.5	15.5	15.4	15.7	14.7	13.0	10.4	7.6	7.9	(35)	
		MCDBR	7.7	8.4	14.9	21.8	22.7	22.3	22.6	20.2	17.1	11.3	8.4	7.7	(36)	
	5% WB	MCWBR	7.6	7.2	10.4	13.3	12.9	12.2	11.8	11.1	10.8	8.8	8.2	7.8	(37)	
		MCWBR	7.3	7.4	13.5	19.2	19.5	19.8	19.4	16.9	15.2	10.7	8.2	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.279	0.314	0.334	0.373	0.369	0.358	0.379	0.375	0.345	0.333	0.301	0.263	(40)	
		taud	2.236	2.311	2.374	2.325	2.397	2.448	2.409	2.419	2.492	2.441	2.310	2.170	(41)	
	Ebn at Noon	Ebn at Noon	176	227	258	264	274	279	269	261	253	219	167	140	(42)	
		Edh at Noon	15	22	27	34	34	33	33	31	24	20	14	11	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	109	289	771	1241	1638	1841	1644	1296	882	398	142	75	(44)	
	RadStd	RadStd	20	49	75	155	139	99	209	117	106	63	27	8	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	+0.83	N/A	N/A	N/A	N/A	-227	-319	N/A	N/A	(46)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page